



MIME

The Multipurpose Internet Mail Extensions is an Internet standard that extends email format to support:

- Non-ASCII text
- Attachments
- Multipart messages
 - E.g. HTML + Plain Text versions in a single email
- Metadata of the email

It was originally designed for email (RFC 2045-2049) but is now widely used in web protocols like HTTP, APIs, etc.

Long story short, MIME is the “grand unified theory” behind:

- How email handle attachments
- Why web servers send “Content-Type: text/css”
- Why Operating Systems know “.json” files are “application/json”

RFC 2045-2049

The MIME standard is defined in “RFC 2045-2049”, and is the foundational source for:

- Email formatting
 - How attachments, text encoding, and multipart messages work
- MIME types
 - Used in HTTP, OS file handling, APIs...
 - E.g. "application/json", "text/html"...

RFC	Title & Purpose
RFC 2045	Format of Internet Message Bodies (structure, headers...)
RFC 2046	Media Types (defines "text/*", "image/*", "multipart/*"...)
RFC 2047	Message Header Extensions (non-ASCII text in headers)
RFC 2048	Registration Procedures (how new MIME Types are added)
RFC 2049	Compliance Criteria (how to implement MIME correctly)

MIME Types

While MIME in email defines how the content is structured (e.g. headers like "Content-Type: text/html"), and allows emails to carry attachments, MIME Types identify file formats in different contexts:

- Operating Systems
 - E.g. the "file" command uses "/etc/mime.types" on Linux
- Web Servers
 - HTTP "Content-Type" header tells web browsers how to handle files
- APIs
 - E.g. JSON responses use "application/json"
- For example, a ".png" file has "image/png" MIME Type

MIME Types are related to the Multipurpose Internet Mail Extensions:

- In email, MIME Types describe parts of the message
 - In this case, MIME focuses on message structure and encoding
- In OS/web, they describe file formats
 - In this case, MIME focuses on file format identification
- Both contexts share the same syntax
 - The "type/subtype" format (e.g. "image/jpeg")